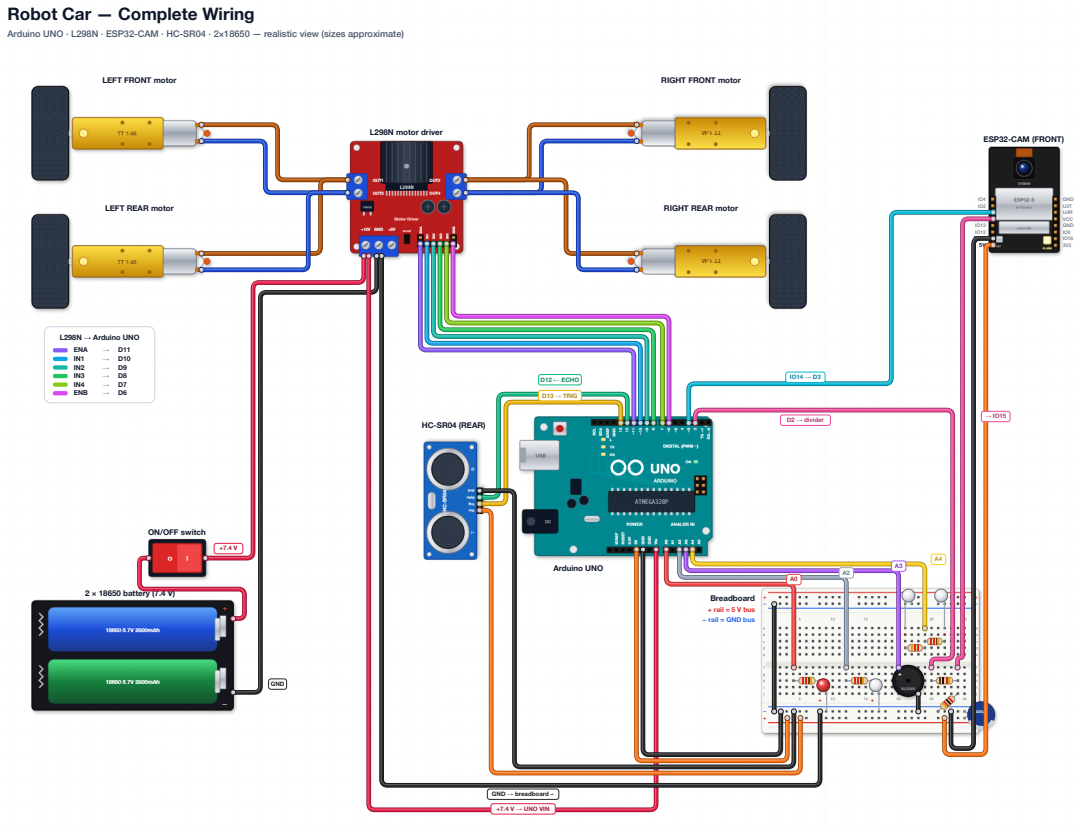
ROBOT CAR PROJECT

MODULE 02

02

Wiring Guide

Every connection step by step, with realistic pictures, connection tables and checks.



Name _____ Class / Group _____ Date _____

4WD · Arduino UNO · ESP32-CAM
Edition 2026

MODULE 02

1 Wiring guide

This module shows **every connection** in the car, one step at a time. Each step has a close-up picture with the wires of that step highlighted, a list of actions and a connection table. Do the steps **in order** and test as you go.

LEARNING OBJECTIVES · ABOUT 120 MIN

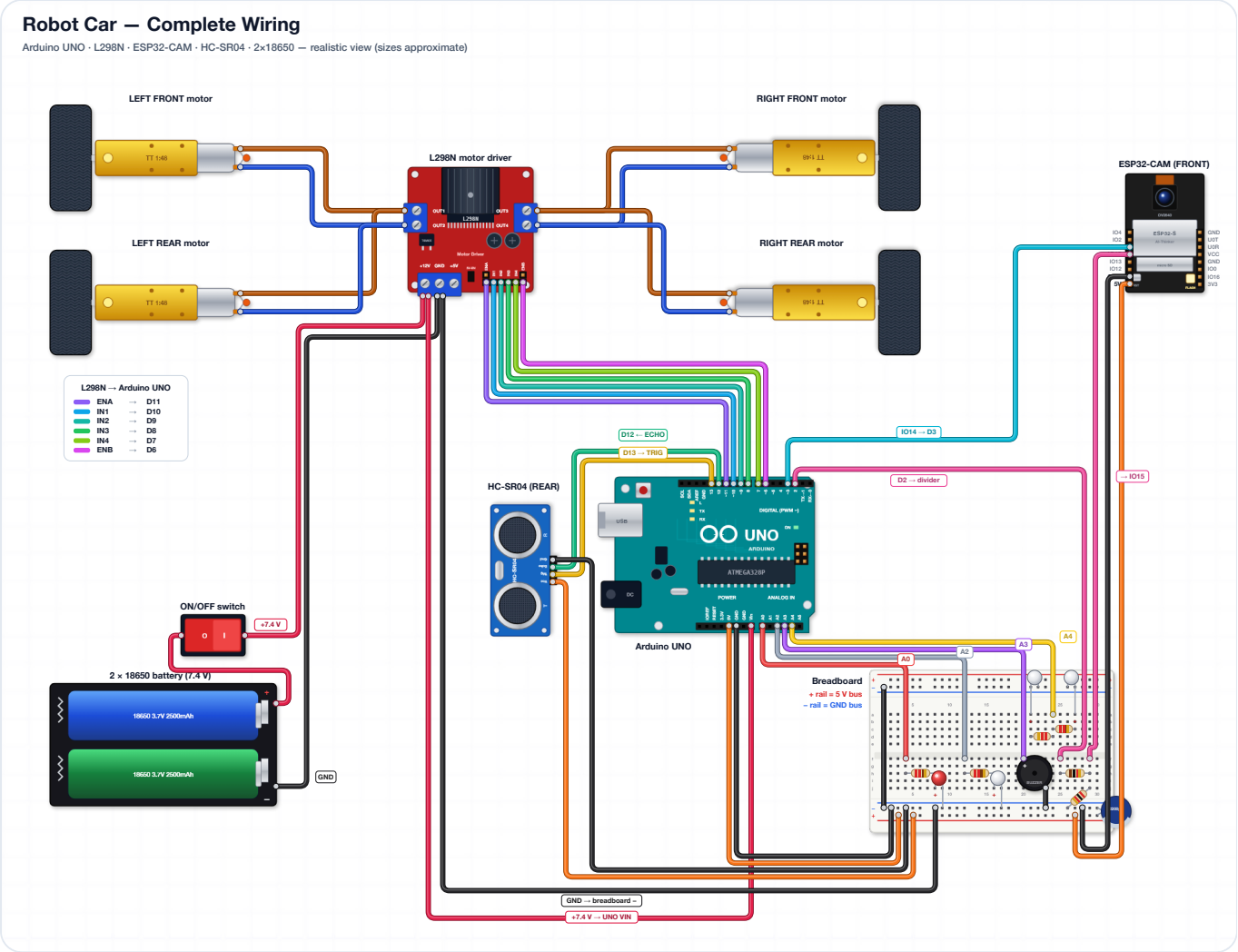
By the end of this module you can:

- Wire a two-rail power system safely (7.4 V and 5 V) with a common ground
- Connect the motors, motor driver, sensor, buzzer and LEDs to the Arduino
- Use a voltage divider to connect a 5 V board to a 3.3 V board
- Check a circuit with a multimeter before powering it

Golden rules: read before you touch a wire

- Always wire with the **battery switched OFF** (better: batteries out of the holder).
- Check the **battery polarity with the multimeter** before the first power-on: battery + must reach the switch, the L298N +12V and the UNO **VIN**.
- **All GNDs are connected together**: battery −, L298N, UNO, ESP32-CAM, sensor, buzzer, LEDs.
- 7.4 V goes **only** to the L298N +12V terminal and the UNO **VIN** pin. Never to a 5V pin or to the ESP32-CAM.
- Never connect a 5 V output straight to an ESP32 pin: use the voltage divider (step 7).

The complete wiring



Complete realistic wiring diagram. An A3 poster version is in **Robot_Car_Wiring_Poster.pdf**.

How to read the pictures

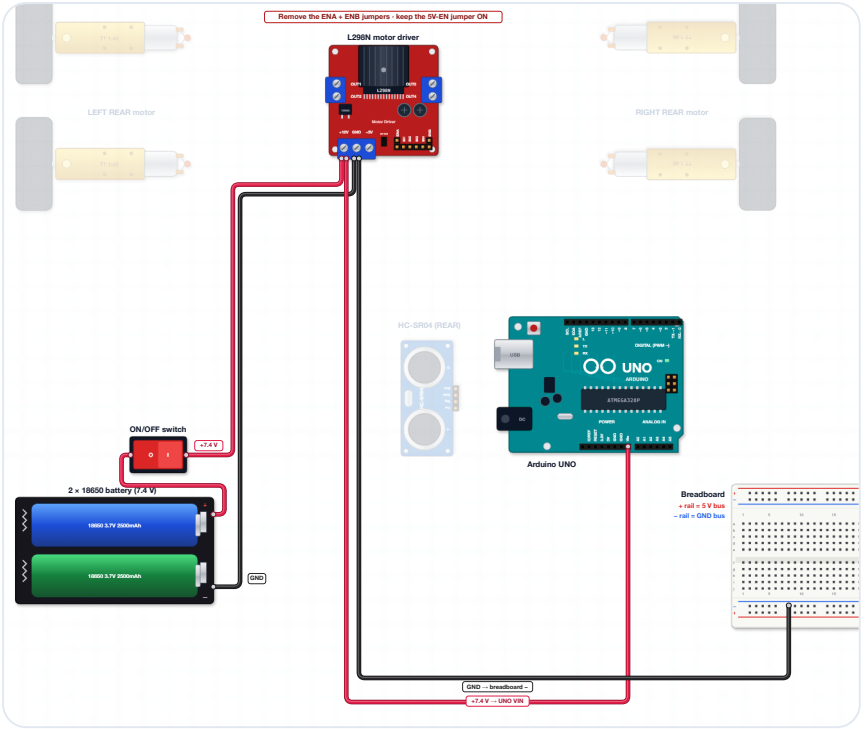
Each coloured line is one jumper wire. Where wires cross they are **not** connected (they are insulated). The breadboard's long **red + rail** is the 5 V bus and the **blue − rail** is the GND bus.

Breadboard rails

On some breadboards the rails are split in the middle. Check with the multimeter (continuity mode) that column 1 and column 30 of each rail are connected. If not, add a short jumper across the gap.

1 Power system

Battery → switch → L298N +12V · second wires: +12V → UNO VIN, GND → breadboard − rail



Two wires in one screw terminal

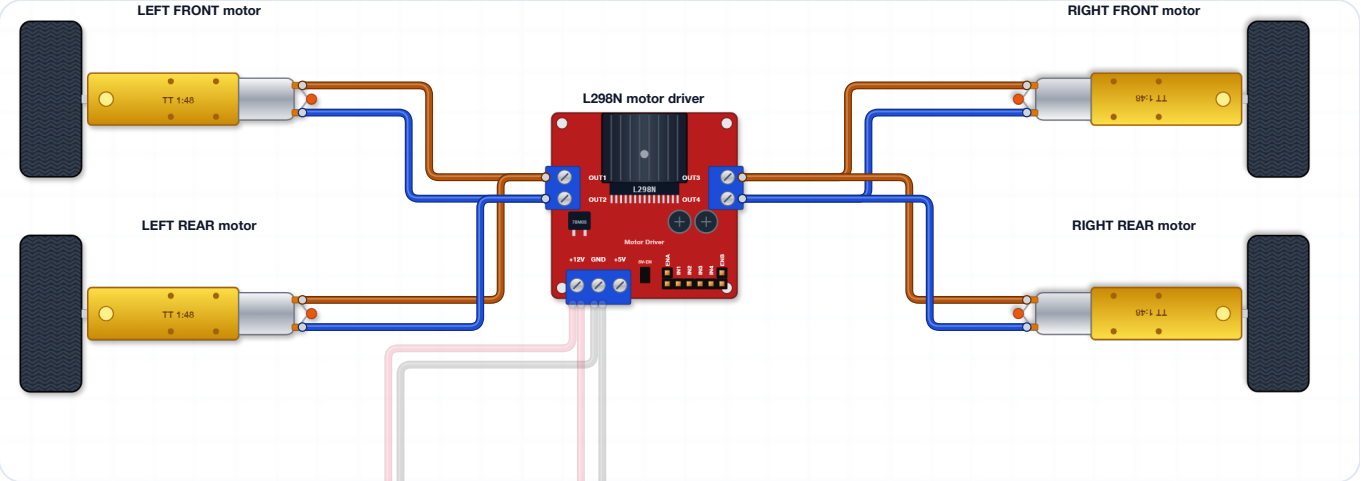
The L298N **+12V** terminal holds two red wires: the one from the switch and the one to the UNO **VIN**. The **GND** terminal holds two black wires: battery − and the wire to the breadboard − rail. Twist the bare ends together, push them in and tighten firmly. No buck converter is needed: the UNO makes the 5 V (step 4).

1. Battery **+** (red) → switch, first pin.
2. Switch second pin → **L298N +12V**.
3. Battery **−** (black) → **L298N GND**.
4. Second red wire: **L298N +12V** → **UNO VIN** (power header, next to GND).
5. Second black wire: **L298N GND** → **breadboard − rail**.
6. On the L298N: **remove the ENA and ENB jumpers** and keep the 5V-EN jumper on. Leave the L298N +5V terminal empty.

	From	To	Note
	Battery +	Switch	22 AWG wire
	Switch	L298N +12V	carries the motor current
	Battery −	L298N GND	
	L298N +12V	`VIN`	7.4 V into the UNO regulator
	L298N GND	Breadboard − rail	= GND bus

2 Motors

Left pair on OUT1/OUT2 · right pair on OUT3/OUT4



The two motors on the same side are wired **in parallel**: both wires of the front motor and both wires of the rear motor go into the same two screws.

1. Solder about 15 cm wires and a **100 nF capacitor** to each motor (see Module 03).
2. Left front + left rear: wire A → **OUT1**, wire B → **OUT2**.
3. Right front + right rear: wire A → **OUT3**, wire B → **OUT4**.
4. Tighten the screws and pull gently on each wire to check it is held.

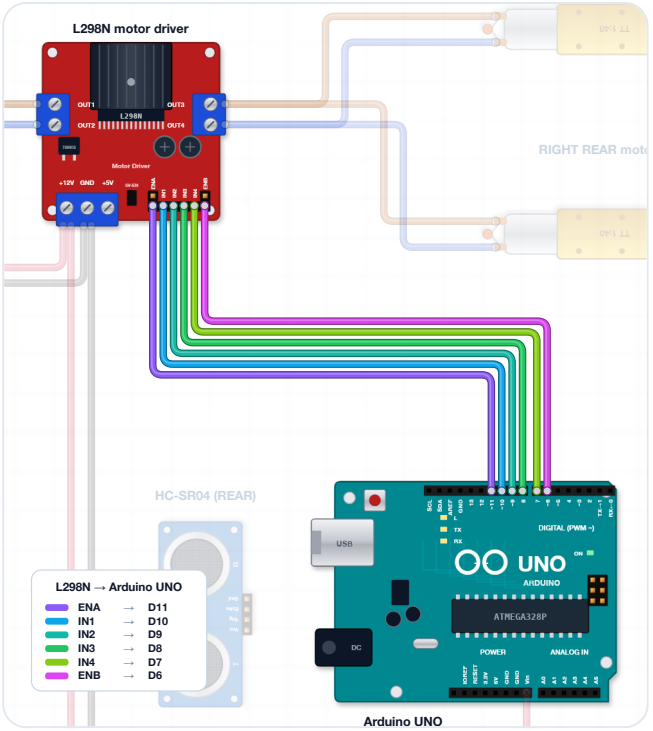
	From	To	Note
	Left motors, wire A	L298N OUT1	both motors
	Left motors, wire B	L298N OUT2	both motors
	Right motors, wire A	L298N OUT3	both motors
	Right motors, wire B	L298N OUT4	both motors

Wrong direction later?

If one motor turns against its partner, swap **only that motor's** two wires. If a whole side is backwards, change `INVERT_LEFT` / `INVERT_RIGHT` in the UNO code.

3 L298N → Arduino UNO

Six control wires, side by side



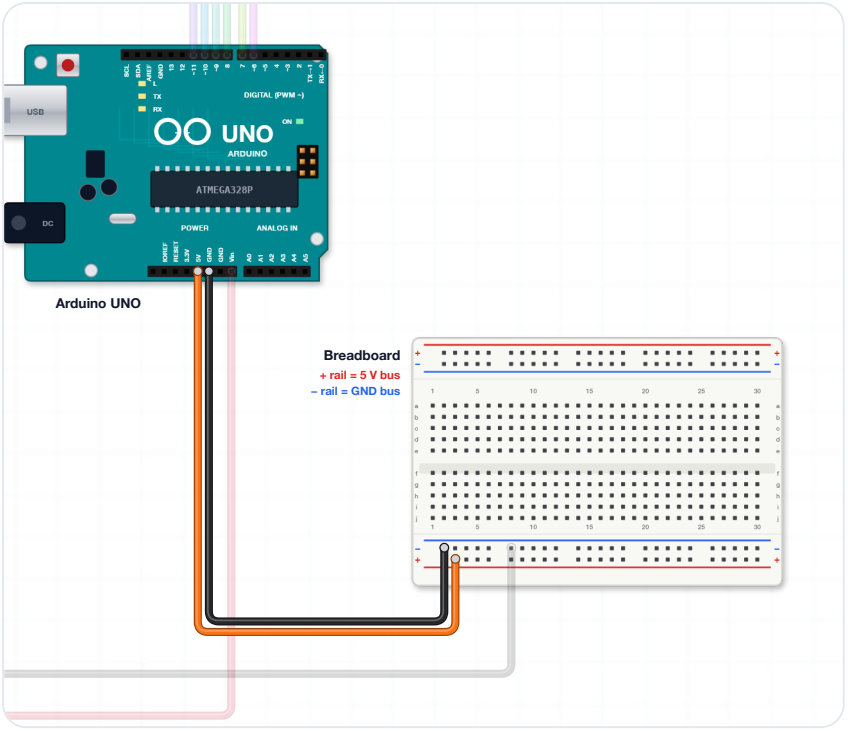
The pins were chosen so the six wires run in the same order on both boards. Use six different colours.

1. ENA → **D11** (PWM ~, left speed).
2. IN1 → **D10**, IN2 → **D9** (left direction).
3. IN3 → **D8**, IN4 → **D7** (right direction).
4. ENB → **D6** (PWM ~, right speed).
5. Test now: upload `tests/01_motor_test` with the wheels in the air (Module 05).

	From	To	Note
	`ENA`	`D11 ~`	left speed (PWM)
	`IN1`	`D10`	left direction
	`IN2`	`D9`	left direction
	`IN3`	`D8`	right direction
	`IN4`	`D7`	right direction
	`ENB`	`D6 ~`	right speed (PWM)

4 5 V rail from the Arduino

UNO 5V and GND to the breadboard rails



The UNO turns the 7.4 V on **VIN** into a steady **5 V**. Its **5V pin** now feeds the breadboard + rail, and that rail powers the ESP32-CAM, the sensor, the buzzer and the LEDs.

1. UNO **GND** → breadboard **– rail**.
2. UNO **5V** → breadboard **+ rail**.
3. Do **not** also plug a power supply into the UNO's DC jack: VIN is already used.
4. Check: battery on, multimeter between the + and – rails = **4.9–5.1 V**.

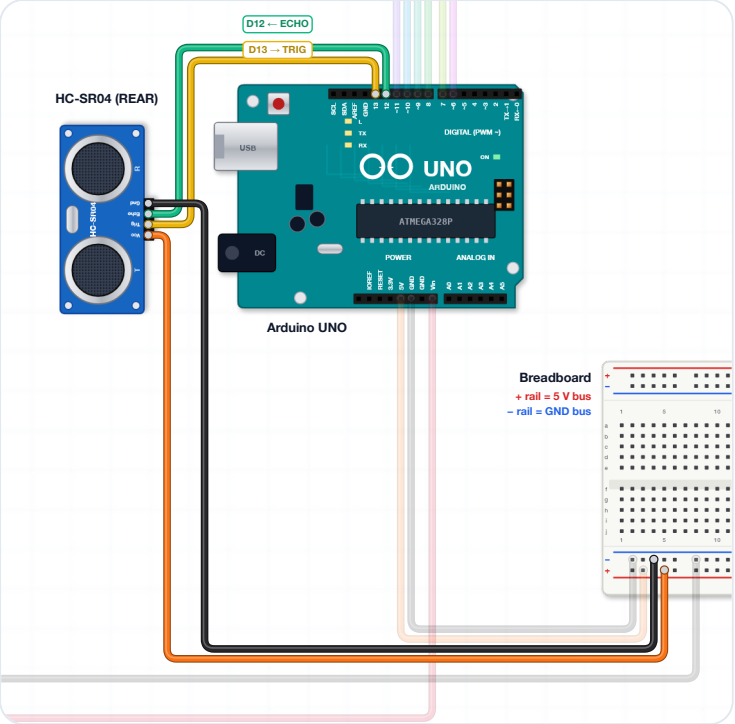
	From	To	Note
▬	`GND`	Breadboard – rail	power header
▬	`5V`	Breadboard + rail	= 5 V bus for everything

Can I keep the USB cable plugged in?

Yes. The UNO picks the stronger supply by itself. On USB only (battery off) the electronics and the camera work but the motors do not move. That is handy while testing.

5 Rear ultrasonic sensor

TRIG → D13 · ECHO → D12 · power from the rails



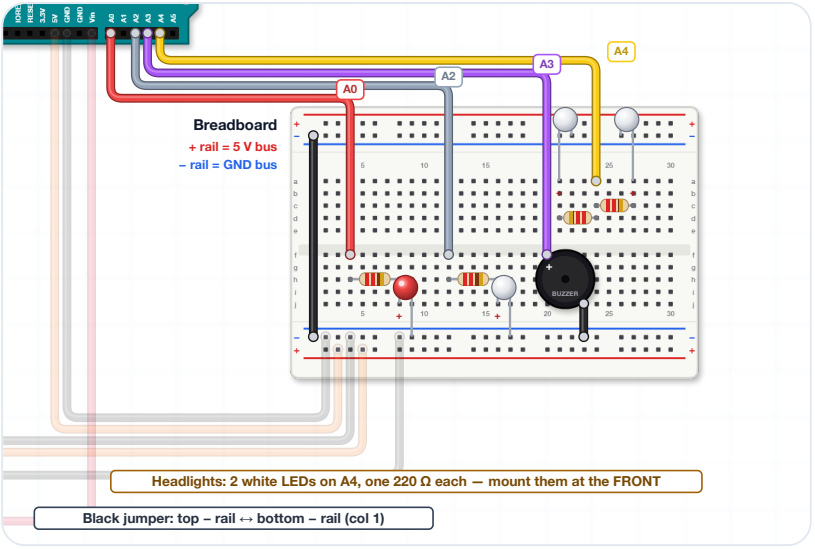
The HC-SR04 is mounted at the **rear** of the car, facing backwards (Module 03 shows the height).

1. HC-SR04 **TRIG** → UNO **D13**.
2. HC-SR04 **ECHO** → UNO **D12**.
3. **VCC** → + rail, **GND** → – rail.
4. Test: `tests/02_ultrasonic_test`.

	From	To	Note
	HC-SR04 TRIG	D13	UNO sends the ping
	HC-SR04 ECHO	D12	UNO measures the echo
	HC-SR04 VCC	+ rail	5 V
	HC-SR04 GND	– rail	

6 Lights + buzzer

A0 red rear LED · A2 white reverse LED · A3 buzzer · A4 two headlights



These parts sit on the breadboard. The analog pins A0–A4 work as normal digital outputs. The rear parts use the **bottom half** of the breadboard, the headlights the **top half**.

1. A0 wire → hole **4f**. 220 Ω resistor from **4h** to **8h**. Red LED: long leg (+) in **8j**, short leg in the **– rail**.
2. A2 wire → **12f**. 220 Ω from **12h** to **16h**. White LED: long leg in **16j**, short leg in the – rail.
3. A3 wire → **20f**. Buzzer **+** leg in **20h**, – leg in **23h**, short black jumper from **23j** to the – rail.
4. **Headlights:** A4 wire → **24a**. 220 Ω from **24c** to **27c**, white LED long leg in **27a**, short leg in the **top – rail**. Second 220 Ω from **24d** to **21d**, white LED long leg in **21a**, short leg in the top – rail.
5. Black jumper from the **top – rail** to the **bottom – rail** (column 1), so the top rail is GND too.
6. On the car the headlights sit at the **front**: put each LED on two female-male jumpers instead of in the breadboard.
7. Test: `tests/03_buzzer_led_test`.

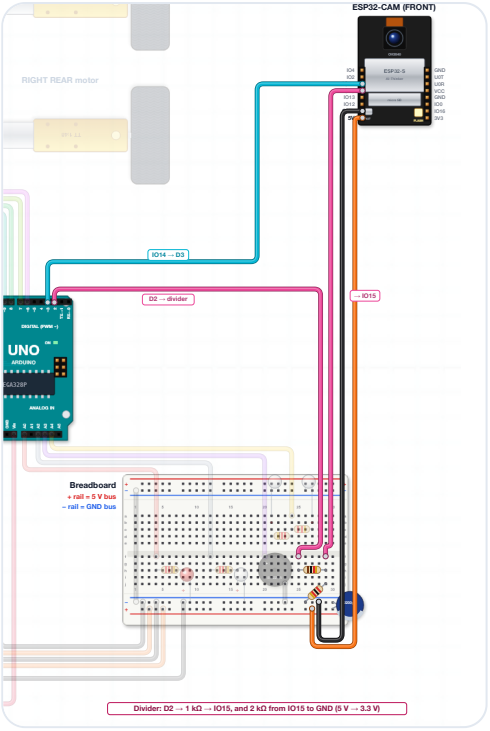
	From	To	Note
	`A0`	220 Ω → red LED (+)	rear warning light
	`A2`	220 Ω → white LED (+)	reverse light
	`A3`	Buzzer +	buzzer – to – rail
	`A4`	2 × (220 Ω → white LED +)	headlights, one resistor each
	Top – rail	Bottom – rail	bridge jumper

LED tips

The **longer leg** is + (anode); an LED wired backwards simply stays off. Each headlight has **its own 220 Ω**: with one shared resistor the brighter LED would take most of the current. Two headlights on A4 (≈ 26 mA) are fine for one UNO pin.

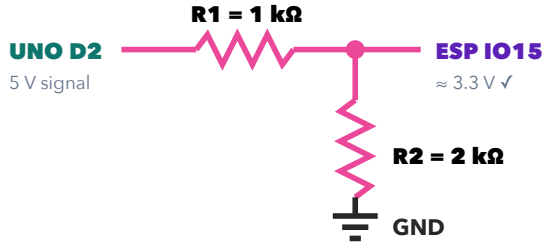
7 ESP32-CAM

IO14 → D3 · D2 → divider → IO15 · 5V / GND · 2200 µF capacitor



Why a voltage divider?

The UNO speaks 5 V, but ESP32 pins only accept 3.3 V. Two resistors make 5 V × 2k / (1k + 2k) = **3.33 V**. The other direction (3.3 V into the UNO) needs no divider: the UNO reads 3.3 V as HIGH.



1. Divider: UNO **D2** wire → hole **25f**. **1 kΩ** from **25h** to **29h**. **2 kΩ** from **29j** to the − rail.
2. ESP32-CAM **IO15** → hole **29f** (middle of the divider).
3. ESP32-CAM **IO14** → UNO **D3** (direct wire).
4. ESP32-CAM **5V** → + rail, **GND** → − rail (use the 5V pin, **not** 3V3).
5. **2200 µF capacitor** across the rails next to the ESP wires: the leg on the side with the stripe (−) goes in the − rail.
6. Leave **U0T, U0R and IO0** free: they are for programming.

	From	To	Note
	`IO14` (TX)	`D3` (RX)	direct wire
	`D2` (TX)	1 kΩ → `IO15` (RX)	2 kΩ from IO15 to GND
	`5V`	+ rail	not 3V3!
	`GND`	− rail	

MODULE 02

Full pin map & final check

Arduino UNO

Pin	Connected to
D2	→ 1 kΩ → ESP32 IO15 (2 kΩ to GND)
D3	← ESP32 IO14
D6 ~	L298N ENB
D7	L298N IN4
D8	L298N IN3
D9	L298N IN2
D10	L298N IN1
D11 ~	L298N ENA
D12	HC-SR04 ECHO
D13	HC-SR04 TRIG
A0	Red rear LED (220 Ω)
A1	Servo (optional)
A2	White reverse LED (220 Ω)
A3	Active buzzer +
A4	2 headlights (220 Ω each)
VIN	L298N +12V (7.4 V)
5V / GND	+ rail / − rail
D0 / D1	free (USB)

ESP32-CAM

Pin	Connected to
5V	+ rail
GND	− rail
IO14	TX → UNO D3
IO15	RX ← divider ← UNO D2
IO4	on-board flash LED = cam light
IO33	on-board red LED = status
U0T U0R IO0	programming only

L298N

Terminal	Connected to
OUT1/OUT2	left motors (parallel)
OUT3/OUT4	right motors (parallel)
+12V	switch (battery +) and UNO VIN
GND	battery − and breadboard − rail
+5V	not connected
ENA…ENB	UNO D11, D10, D9, D8, D7, D6

Final check before the first power-on

- [ ] Battery polarity checked, 5 V rail reads 4.9–5.1 V
- [ ] ENA/ENB jumpers removed, 5V-EN jumper on
- [ ] No 7.4 V wire touches a 5 V or signal pin
- [ ] All grounds connected (check with the multimeter)
- [ ] Divider in place between D2 and IO15
- [ ] ESP32-CAM on 5V (not 3V3)
- [ ] 2200 µF capacitor polarity correct
- [ ] Wheels off the ground for the first test

Wired by (student) _____ Checked by (teacher) _____ Date _____

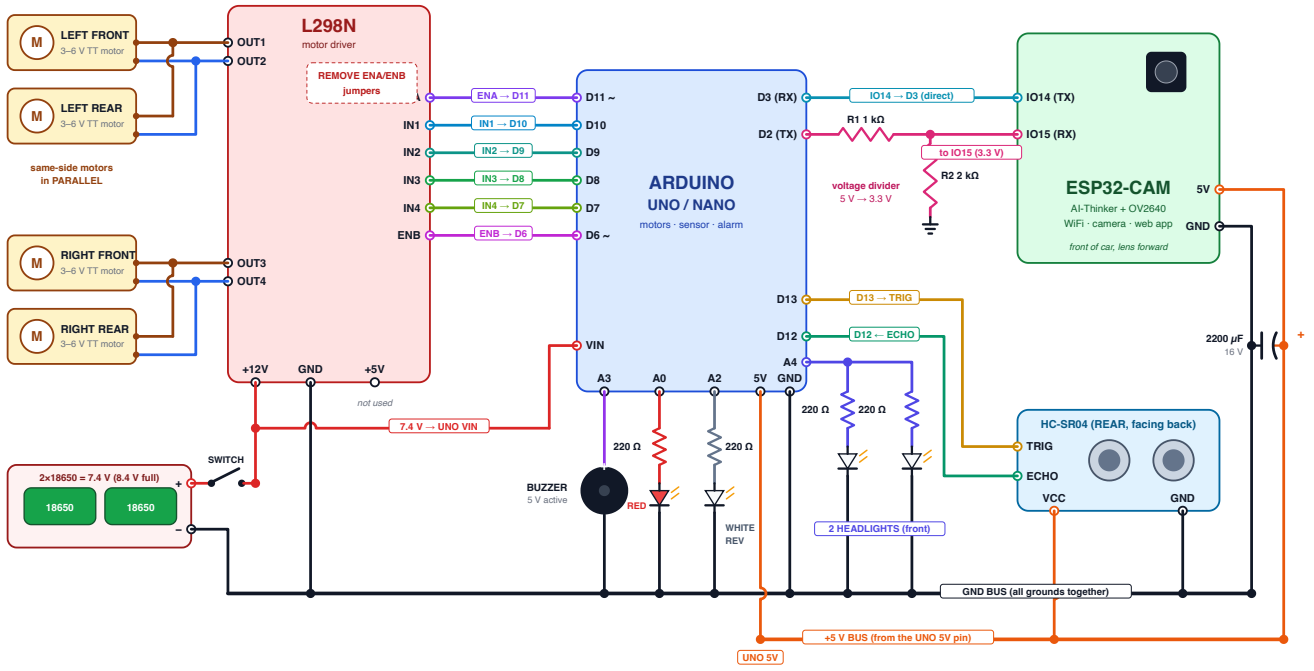
MODULE 02

Schematic view

The same circuit drawn as a schematic. It is useful for understanding and for finding faults.

ROBOT CAR – SCHEMATIC WIRING DIAGRAM

4WD chassis · L298N · Arduino UNO/Nano · ESP32-CAM · HC-SR04 (rear) · buzzer + rear LEDs · 2 headlights · 2×18650 (no buck converter: the UNO makes the 5 V)



POWER — no buck converter needed

- Battery 7.4 V → switch → L298N +12V. A 2nd wire in the same +12V terminal → UNO VIN.
- The UNO regulator makes the 5 V: UNO 5V pin → 5 V bus (ESP32-CAM, HC-SR04, buzzer, LEDs).
- 2200 µF 16 V capacitor right at the ESP32-CAM 5V/GND pins (+ leg to 5V).
- L298N: keep the 5V-EN jumper ON, remove ENA/ENB jumpers, leave its +5V pin unused.
- Do not add a servo to this 5 V bus (the UNO regulator is too small for it).

WIRE COLOURS

Battery +7.4 V · +5 V · GND · Motor wires · UART ESP→UNO
UART UNO→ESP · TRIG · ECHO · Buzzer · Headlights

- Wires crossing with a small hop are NOT connected. A black dot = connected.
- **ALL grounds must be common. Never feed 7.4 V into the ESP32-CAM or UNO 5V pin.**